

Lesson 12: Introduction to the Calculus of Variations

ELTE Eötvös Loránd University
Faculty of Informatics, Department of Numerical Analysis

2025

Contents

1	Introduction: From Functions to Functionals	2
1.1	A Three-Way Comparison	2
2	Functionals and Their Extrema	2
2.1	What Is a Functional?	2
2.2	A First Example: Area Under a Curve	3
3	Variational Problems: Physical Motivation	4
3.1	The Hanging Rope: Minimising Gravitational Potential Energy	4
3.2	The Shortest Path Between Two Points	4
3.3	The General Variational Problem	5
4	The First Variation and the Fundamental Lemma	5
4.1	The Variation of a Function	5
4.2	The First Variation	6
4.3	The Fundamental Lemma of the Calculus of Variations	7
5	The Euler–Lagrange Equation	7
5.1	The Necessary Condition for an Extremal	7
5.2	Worked Examples	8
6	Constrained Extrema of Functions	10
6.1	The Classical Lagrange Multiplier Method	10
6.2	Several Constraints	11
6.3	Connection with Variational Problems	11
7	Constrained Variational Problems and Lagrange Multipliers	11
7.1	Integral Constraints	11
7.2	The Multiplier Principle for Functionals	12
7.3	The Beltrami Identity	13
7.4	Application: The Catenary	13
7.5	Practical Recipe	15

1 Introduction: From Functions to Functionals

1.1 A Three-Way Comparison

Classical analysis teaches us to find the minimum or maximum of a function. The simplest setting is a real-valued function of one real variable, $f(x) : \mathbb{R} \rightarrow \mathbb{R}$. To locate a candidate extremum we solve the *necessary condition* $f'(x) = 0$; any solution is a *stationary point* and a candidate for a local minimum, maximum, or inflection point. The *sufficient condition* then settles the question: if $f''(x) > 0$ at a stationary point, that point is a local minimum; if $f''(x) < 0$, it is a local maximum; if $f''(x) = 0$, higher-order analysis is needed.

Moving one step further, consider a function of several variables, $f(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}$. The gradient now plays the role of the derivative: the necessary condition is $\nabla f = \mathbf{0}$, meaning all partial derivatives vanish simultaneously. The second-order test is conducted via the *Hessian matrix* H of second partial derivatives: a positive definite H indicates a local minimum, a negative definite H a local maximum, and an indefinite H a saddle point.

Calculus of variations represents an entirely new dimension: instead of optimising over numbers or finite-dimensional vectors, we optimise over *functions*. We are given a **functional**

$$J[y] = \int_a^b F(x, y(x), y'(x)) dx,$$

where F is a prescribed function of three arguments, and we seek the function $y(x)$ from some admissible class that renders $J[y]$ stationary (minimal, maximal, or saddle). The necessary condition is written $\delta J = 0$, where δJ is called the *first variation* of the functional. This condition leads to the celebrated **Euler–Lagrange equation**:

$$\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) = 0.$$

A solution $Y(x)$ of this ordinary differential equation is called an *extremal* of the functional J . The parallel with classical analysis is striking and deep: the role of $f'(x) = 0$ is played by the Euler–Lagrange equation; the role of f'' is played by the second variation $\delta^2 J$, which we will not pursue further in this introduction.

The progression from a single real number, to a finite-dimensional vector, to a function is the conceptual backbone of this lecture. It is remarkable that the step from $\nabla f = \mathbf{0}$ to the Euler–Lagrange equation requires genuinely new analytical machinery: the notion of a functional, the concept of a variation, and a key auxiliary result known as the Fundamental Lemma. We now build these ideas from the ground up.

2 Functionals and Their Extrema

2.1 What Is a Functional?

Definition 2.1 (Functional). A **functional** is a mapping

$$J[y] : S \longrightarrow \mathbb{R},$$

where the domain S is a *set of functions*. That is, a functional takes a *function* as its input and returns a *real number* as its output.

It is worth pausing to appreciate exactly how different this is from an ordinary function. When we write $f(x) = x^2$, the argument x is a real number. When we write

$$J[y] = \int_0^1 y(x) dx,$$

the argument y is itself a function — a curve, a shape, a trajectory. Two different functions y_1 and y_2 will, in general, give two different numbers $J[y_1]$ and $J[y_2]$. The square bracket notation $J[\cdot]$ (as opposed to the round bracket $f(\cdot)$) is a conventional reminder that the argument is a function, not a number.

Definition 2.2 (Extremum of a Functional). We say that the functional $J[y]$ has an **extremum** at $y^* \in S$ if $J[y^*]$ attains the maximum or minimum value of J over the entire admissible class S .

2.2 A First Example: Area Under a Curve

Let $\mathcal{C}[0, 1]$ be the set of all continuous functions on $[0, 1]$, and define

$$J[y] = \int_0^1 y(x) dx.$$

This functional simply computes the area under the graph of y from $x = 0$ to $x = 1$. For any continuous y , this integral is a well-defined real number, so $J : \mathcal{C}[0, 1] \rightarrow \mathbb{R}$ is a perfectly legitimate functional.

To develop intuition, we evaluate J on a selection of familiar functions:

Function $y(x)$	$J[y] = \int_0^1 y(x) dx$
x	0.5
x^2	0.333
x^3	0.25
$\sin x$	0.4597
$\cos x$	0.8415
e^x	1.7183
1	1

Among these seven, the smallest value is attained at $y(x) = x^3$ (giving $J = 0.25$) and the largest at $y(x) = e^x$ (giving $J \approx 1.718$). This is already interesting: the functional J assigns a distinct number to each function, and that number depends on the *global* shape of y over the whole interval $[0, 1]$, not merely on its value at a single point. Changing the function even slightly will, in general, change the value of J . This global dependence is what makes the optimisation of functionals conceptually richer than the optimisation of ordinary functions.

Remark 2.1. Over the unrestricted class $\mathcal{C}[0, 1]$, the functional $J[y] = \int_0^1 y(x) dx$ has no finite supremum (take $y_M(x) = M$ for any large constant M ; then $J[y_M] = M \rightarrow +\infty$). Interesting extremum problems arise when we restrict the admissible class S , for example by fixing the values of y at the endpoints or by prescribing the total length of the curve. Such constraints are exactly what the variational problems in the next section impose.

3 Variational Problems: Physical Motivation

Before developing the general theory, it is instructive to see where variational problems arise naturally in physics and geometry. The two examples below have shaped the history of the subject and guided the development of the mathematical tools we will introduce.

3.1 The Hanging Rope: Minimising Gravitational Potential Energy

Consider a flexible, inextensible rope of total mass M and total length L , suspended from two fixed endpoints. Observation and physical intuition agree that the rope hangs in a smooth, drooping curve. The precise question of *which* curve is one of the oldest problems in variational calculus: the rope settles into the shape that minimises its gravitational potential energy.

We model the rope as a curve $y = y(x)$ over a horizontal interval $[0, L]$ and derive the functional for the total potential energy. Divide the rope into n thin segments. The i -th segment spans a horizontal distance Δx and has a height difference Δy . By the Pythagorean theorem, the arc length of this segment is approximately

$$s_i \approx \sqrt{(\Delta x)^2 + (\Delta y)^2} = \Delta x \cdot \sqrt{1 + \left(\frac{\Delta y}{\Delta x}\right)^2}.$$

Since the rope is uniform (mass is distributed proportionally to length), the mass of the i -th segment is

$$m_i = \frac{M}{L} s_i \approx \frac{M}{L} \sqrt{1 + (y'_i)^2} \Delta x,$$

where $y'_i = \Delta y_i / \Delta x$ denotes the slope of that segment. Treating the segment as a point mass located at height $y(x_i)$, its contribution to the gravitational potential energy is

$$\Delta E_{p,i} = g y(x_i) m_i \approx \frac{Mg}{L} y(x_i) \sqrt{1 + (y'_i)^2} \Delta x.$$

Summing over all n segments and taking the limit as $n \rightarrow \infty$ (and $\Delta x \rightarrow 0$), the sum becomes an integral and we arrive at the **gravitational potential energy functional**:

$$E_p[y] = \frac{Mg}{L} \int_0^L y(x) \sqrt{1 + (y'(x))^2} dx.$$

The physical equilibrium corresponds to the function $y(x)$ that *minimises* $E_p[y]$, subject to the constraints that the rope's endpoints are fixed and its total arc length equals L . This constrained minimisation problem is a variational problem of the type $J[y] = \int F(x, y, y') dx$ (with $F = (Mg/L) y \sqrt{1 + (y')^2}$), and its solution is the celebrated **catenary**, $y(x) = a \cosh(x/a) + b$.

3.2 The Shortest Path Between Two Points

Given two points $P_a = (a, y_1)$ and $P_b = (b, y_2)$ in the plane, which smooth curve $y(x)$ connecting them has the shortest total arc length? The answer is of course a straight line, but variational calculus will derive this fact systematically.

The arc length element of a smooth curve is obtained by the same Pythagorean argument as above:

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + \left(\frac{dy}{dx}\right)^2} dx = \sqrt{1 + (y')^2} dx.$$

Integrating from $x = a$ to $x = b$, we obtain the **arc length functional**:

$$S[y] = \int_a^b \sqrt{1 + (y')^2} dx.$$

The problem is to find the function $y(x)$ with $y(a) = y_1$, $y(b) = y_2$ that *minimises* $S[y]$. Here $F(x, y, p) = \sqrt{1 + p^2}$ (independent of x and y). We will solve this explicitly in the next section using the Euler–Lagrange equation.

3.3 The General Variational Problem

Both preceding examples are special cases of the following standard framework.

Definition 3.1 (Standard Variational Problem). Given a function $F(x, y, p)$ that is twice continuously differentiable in all three arguments, the **standard variational problem** is:

$$\text{find } y^*(x) \text{ that extremises } J[y] = \int_{x_1}^{x_2} F(x, y(x), y'(x)) dx$$

subject to the **boundary conditions**

$$y(x_1) = y_1, \quad y(x_2) = y_2.$$

A function $Y(x)$ at which J achieves its minimum or maximum value is called an **extremal** of the functional.

The function F is often called the **Lagrangian** of the problem, in analogy with mechanics, where the Lagrangian encodes the dynamics of a system. Different choices of F yield entirely different physical or geometric problems, yet the mathematical framework for finding extremals is always the same: derive and solve the Euler–Lagrange equation.

4 The First Variation and the Fundamental Lemma

4.1 The Variation of a Function

Suppose $Y(x)$ is a candidate extremal. How should we test whether it truly minimises or maximises J ? The key idea, due to Euler and Lagrange, is to compare Y against a family of nearby competitor functions obtained by adding a small *perturbation*.

Definition 4.1 (Variation of a Function). Let $Y(x)$ be a smooth function on $[x_1, x_2]$ satisfying $Y(x_1) = y_1$ and $Y(x_2) = y_2$. A **variation** of Y is any one-parameter family of the form

$$y_\varepsilon(x) = Y(x) + \varepsilon g(x),$$

where:

- $\varepsilon \in \mathbb{R}$ is a small real parameter,

- $g(x)$ is a smooth function satisfying $g(x_1) = g(x_2) = 0$.

The condition $g(x_1) = g(x_2) = 0$ ensures that every competitor y_ε satisfies the same boundary conditions as Y , regardless of the choice of g . The function g is sometimes written $\delta y = \varepsilon g$ and called the *virtual displacement*: it represents an infinitesimal change of shape that we are “testing” against Y .

4.2 The First Variation

Evaluating the functional on a variation yields an ordinary function of the parameter ε :

$$\Phi(\varepsilon) = J[Y + \varepsilon g] = \int_{x_1}^{x_2} F(x, Y(x) + \varepsilon g(x), Y'(x) + \varepsilon g'(x)) dx.$$

If Y is an extremal, then $\Phi(\varepsilon)$ has an extremum at $\varepsilon = 0$ (for each admissible g). A necessary condition is therefore $\Phi'(0) = 0$.

Definition 4.2 (First Variation). The **first variation** of J at Y in the direction g is

$$\delta J[Y; g] = \left. \frac{d}{d\varepsilon} J[Y + \varepsilon g] \right|_{\varepsilon=0}.$$

This is also called the **Gâteaux derivative** of the functional J at Y in the direction g . Let us compute it explicitly. Differentiating under the integral sign:

$$\begin{aligned} \delta J[Y; g] &= \int_{x_1}^{x_2} \left. \frac{\partial}{\partial \varepsilon} F(x, Y + \varepsilon g, Y' + \varepsilon g') \right|_{\varepsilon=0} dx \\ &= \int_{x_1}^{x_2} \left[\frac{\partial F}{\partial y} g(x) + \frac{\partial F}{\partial y'} g'(x) \right] dx. \end{aligned}$$

We integrate the second term by parts (with $u = \partial F / \partial y'$ and $dv = g' dx$):

$$\int_{x_1}^{x_2} \frac{\partial F}{\partial y'} g'(x) dx = \left[\frac{\partial F}{\partial y'} g(x) \right]_{x_1}^{x_2} - \int_{x_1}^{x_2} \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) g(x) dx.$$

The boundary term vanishes because $g(x_1) = g(x_2) = 0$. Therefore, the first variation simplifies to:

$$\delta J[Y; g] = \int_{x_1}^{x_2} \left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right] g(x) dx.$$

Definition 4.3 (Stationary Function). A function $Y(x)$ is called a **stationary function** of the functional J if

$$\delta J[Y; g] = 0 \quad \text{for all smooth } g \text{ with } g(x_1) = g(x_2) = 0.$$

Every extremal is a stationary function: if $J[Y] \leq J[Y + \varepsilon g]$ (or $\geq$) for all small ε and all admissible g , then $\Phi'(0) = \delta J[Y; g] = 0$ for each such g .

4.3 The Fundamental Lemma of the Calculus of Variations

We have shown that if Y is a stationary function, then

$$\int_{x_1}^{x_2} h(x) g(x) dx = 0 \quad \text{for all smooth } g \text{ with } g(x_1) = g(x_2) = 0,$$

where we write $h(x) = \left. \frac{\partial F}{\partial y} \right|_Y - \frac{d}{dx} \left(\left. \frac{\partial F}{\partial y'} \right|_Y \right)$ for brevity. The question is: what can we conclude about h from this?

Theorem 4.1 (Fundamental Lemma of the Calculus of Variations). *Let $h(x)$ be continuous on $[a, b]$. If*

$$\int_a^b h(x) g(x) dx = 0$$

for every smooth function $g : [a, b] \rightarrow \mathbb{R}$ satisfying $g(a) = g(b) = 0$, then

$$h(x) \equiv 0 \quad \text{on } [a, b].$$

Proof sketch. Suppose, for contradiction, that h is not identically zero. By continuity, there is some point $x_0 \in (a, b)$ and a sub-interval $(c, d) \subset (a, b)$ with $x_0 \in (c, d)$ on which h has constant sign, say $h(x) > 0$ on (c, d) . Define

$$g(x) = \begin{cases} (x-c)^2(x-d)^2 & x \in [c, d], \\ 0 & \text{otherwise.} \end{cases}$$

Then g is smooth, $g(a) = g(b) = 0$, and $g(x) > 0$ for $x \in (c, d)$. Hence

$$\int_a^b h(x) g(x) dx = \int_c^d h(x) g(x) dx > 0,$$

contradicting the assumption. Therefore $h \equiv 0$. □

The Fundamental Lemma is the essential bridge that allows us to pass from an *integral condition* ($\delta J = 0$ for all g) to a *pointwise differential equation* (the Euler–Lagrange equation). Without it, we would be stuck with a condition involving integrals; with it, we obtain a concrete ODE whose solutions are the extremals.

5 The Euler–Lagrange Equation

5.1 The Necessary Condition for an Extremal

We now have all the ingredients needed for the main result.

Theorem 5.1 (Euler–Lagrange Equation). *Let $F(x, y, p)$ be twice continuously differentiable. Suppose $Y(x)$ is an extremal of the functional*

$$J[y] = \int_{x_1}^{x_2} F(x, y(x), y'(x)) dx, \quad y(x_1) = y_1, \quad y(x_2) = y_2.$$

*Then $Y(x)$ satisfies the **Euler–Lagrange equation**:*

$$\frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) - \frac{\partial F}{\partial y} = 0. \tag{1}$$

Proof. Since Y is an extremal, it is a stationary function, so $\delta J[Y; g] = 0$ for all admissible g . The computation in Section 4.2 gives

$$\int_{x_1}^{x_2} \underbrace{\left[\frac{\partial F}{\partial y} - \frac{d}{dx} \left(\frac{\partial F}{\partial y'} \right) \right]}_{=: h(x)} g(x) dx = 0.$$

The expression $h(x)$ is continuous (since F is twice differentiable and Y is smooth). By Theorem 4.1 (Fundamental Lemma), we conclude $h(x) \equiv 0$, which is exactly (1). $\square$

Remark 5.1. Expanding the total derivative in (1), the Euler–Lagrange equation reads

$$F_{y'y''} y'' + F_{y'y} y' + F_{y'x} - F_y = 0.$$

In general this is a *second-order nonlinear ODE* for $Y(x)$. Its two integration constants are fixed by the boundary conditions $Y(x_1) = y_1$ and $Y(x_2) = y_2$. An extremal is thus a solution to this boundary-value problem.

Remark 5.2. The Euler–Lagrange equation is only a *necessary* condition: every extremal must satisfy it, but not every solution of (1) is necessarily an extremal of J . Solutions of the Euler–Lagrange equation are called **extremals** (or *stationary curves*) regardless of whether they give a minimum, maximum, or neither. Determining which case holds requires the analysis of the second variation $\delta^2 J$, which is beyond the scope of this introductory lesson.

5.2 Worked Examples

We now apply the Euler–Lagrange equation to three concrete problems. In each case the procedure is the same: identify F , compute $\partial F/\partial y$ and $\partial F/\partial y'$, write down the ODE (1), and solve subject to the boundary conditions.

Example 5.1 (Extremals of $\int_{x_0}^{x_1} \frac{y'^2}{x^3} dx$). The Lagrangian is $F(x, y, y') = y'^2/x^3$. We compute the required partial derivatives:

$$\frac{\partial F}{\partial y} = 0, \quad \frac{\partial F}{\partial y'} = \frac{2y'}{x^3}.$$

The Euler–Lagrange equation $\frac{d}{dx}(2y'/x^3) = 0$ expands as follows:

$$\frac{d}{dx} \left(\frac{2y'}{x^3} \right) = \frac{2x^3 y'' - 2y' \cdot 3x^2}{x^6} = \frac{2(xy'' - 3y')}{x^4} = 0,$$

so we need to solve

$$xy'' - 3y' = 0.$$

Writing this as $y''/y' = 3/x$ and integrating both sides with respect to x :

$$\ln |y'| = 3 \ln |x| + \text{const} \implies y' = C_1 x^3.$$

Integrating once more:

$$y(x) = \frac{C_1}{4} x^4 + C_2 = Ax^4 + B,$$

where the constants $A = C_1/4$ and B are determined by the boundary conditions. The family of extremals consists of all quartic polynomials of the form $Ax^4 + B$.

Example 5.2 (Extremise $I(y) = \int_{x_1}^{x_2} (1 + y'^2) dx$). Here $F(x, y, y') = 1 + y'^2$. The partial derivatives are

$$\frac{\partial F}{\partial y} = 0, \quad \frac{\partial F}{\partial y'} = 2y'.$$

The Euler–Lagrange equation becomes

$$\frac{d}{dx}(2y') - 0 = 0 \implies 2y'' = 0 \implies y''(x) = 0.$$

Integrating twice:

$$y'(x) = C, \quad y(x) = Cx + D,$$

so the extremals are *straight lines*. The boundary conditions $y(x_1) = y_1$ and $y(x_2) = y_2$ uniquely determine

$$C = \frac{y_2 - y_1}{x_2 - x_1}, \quad D = y_1 - Cx_1,$$

giving the straight-line segment connecting the two endpoints. This confirms our geometric intuition: among all smooth curves joining (x_1, y_1) to (x_2, y_2) , the straight line is the extremal of the functional $I[y] = \int (1 + y'^2) dx$.

Example 5.3 (Find the extremal of $\int_0^1 (y'^2 + 12xy) dx$ with $y(0) = 0$, $y(1) = 1$). The Lagrangian is $F(x, y, y') = y'^2 + 12xy$. We compute:

$$\frac{\partial F}{\partial y} = 12x, \quad \frac{\partial F}{\partial y'} = 2y'.$$

The Euler–Lagrange equation gives

$$\frac{d}{dx}(2y') - 12x = 0 \implies 2y'' = 12x \implies y'' = 6x.$$

Integrating once: $y'(x) = 3x^2 + C_1$. Integrating again:

$$y(x) = x^3 + C_1x + C_2.$$

We apply the boundary conditions:

$$y(0) = 0 \implies C_2 = 0; \quad y(1) = 1 \implies 1 + C_1 = 1 \implies C_1 = 0.$$

The unique extremal is therefore

$$\boxed{y(x) = x^3}.$$

We can verify this directly: $y(0) = 0$, $y(1) = 1$, and $y'' = 6x$ matches the Euler–Lagrange requirement $y'' = 6x$. ✓

It is instructive to confirm the result by substituting $y = x^3$ into the functional:

$$I[x^3] = \int_0^1 ((3x^2)^2 + 12x \cdot x^3) dx = \int_0^1 (9x^4 + 12x^4) dx = \int_0^1 21x^4 dx = \frac{21}{5}.$$

Any other smooth function satisfying the same boundary conditions gives a value of I that differs from $21/5$ (and in fact is larger, since this extremal is a minimum).

6 Constrained Extrema of Functions

6.1 The Classical Lagrange Multiplier Method

Before passing to constrained variational problems, it is useful to recall the original finite-dimensional Lagrange multiplier method. Here the object being optimised is not a functional, but an ordinary function of several variables. Let

$$f : \mathbb{R}^n \rightarrow \mathbb{R}$$

be the function to be optimised, and suppose the admissible points are restricted by one constraint

$$g(x_1, \dots, x_n) = c.$$

If x^* is a constrained local extremum, then f cannot change to first order along any direction tangent to the constraint surface. Tangent directions are orthogonal to the normal vector $\nabla g(x^*)$. Hence $\nabla f(x^*)$ must also be orthogonal to every tangent direction, so it must be parallel to $\nabla g(x^*)$.

Theorem 6.1 (Lagrange Multiplier Method). *Let $f, g : \mathbb{R}^n \rightarrow \mathbb{R}$ be continuously differentiable. Suppose x^* is a constrained local extremum of f subject to $g(x) = c$. If*

$$\nabla g(x^*) \neq 0,$$

then there exists $\lambda \in \mathbb{R}$ such that

$$\nabla f(x^*) + \lambda \nabla g(x^*) = 0.$$

The sign convention is not important; one often writes the same condition as $\nabla f = \lambda \nabla g$. Computationally, introduce the Lagrange function

$$\mathcal{L}(x, \lambda) = f(x) + \lambda(g(x) - c)$$

and solve

$$\frac{\partial \mathcal{L}}{\partial x_i} = 0 \quad (i = 1, \dots, n), \quad \frac{\partial \mathcal{L}}{\partial \lambda} = g(x) - c = 0.$$

The last equation restores the original constraint.

Example 6.1 (Maximal area rectangle with fixed perimeter). Let a rectangle have side lengths $x > 0$ and $y > 0$. Its area is

$$f(x, y) = xy,$$

and fixing the perimeter gives

$$g(x, y) = 2x + 2y = P.$$

The Lagrange function is

$$\mathcal{L}(x, y, \lambda) = xy + \lambda(2x + 2y - P).$$

The first-order equations are

$$\frac{\partial \mathcal{L}}{\partial x} = y + 2\lambda = 0, \quad \frac{\partial \mathcal{L}}{\partial y} = x + 2\lambda = 0,$$

$$\frac{\partial \mathcal{L}}{\partial \lambda} = 2x + 2y - P = 0.$$

The first two equations give $x = y$, and then the constraint gives

$$4x = P, \quad x = y = \frac{P}{4}.$$

Thus, among rectangles with fixed perimeter, the square has maximal area.

6.2 Several Constraints

With several constraints

$$g_1(x) = c_1, \quad \dots, \quad g_m(x) = c_m,$$

one introduces one multiplier for each constraint:

$$\mathcal{L}(x, \lambda_1, \dots, \lambda_m) = f(x) + \lambda_1(g_1(x) - c_1) + \dots + \lambda_m(g_m(x) - c_m).$$

In the regular case, when the constraint gradients are independent, constrained extrema satisfy

$$\nabla f + \lambda_1 \nabla g_1 + \dots + \lambda_m \nabla g_m = 0$$

together with the original constraints.

6.3 Connection with Variational Problems

The next section applies the same idea in a space of functions. The unknown is no longer a vector $x \in \mathbb{R}^n$, but an entire function $y(x)$. The role of the gradient is played by the first variation, and a single scalar integral constraint again produces a single constant multiplier.

7 Constrained Variational Problems and Lagrange Multipliers

7.1 Integral Constraints

The standard Euler–Lagrange equation applies directly when the only restrictions on the competitors are the endpoint conditions. Many physical problems impose an additional global constraint. The hanging rope is the basic example: the rope may change shape, but its total length is fixed.

Definition 7.1 (Isoperimetric Variational Problem). An **isoperimetric variational problem** asks us to extremise

$$J[y] = \int_a^b F(x, y, y') dx$$

subject to the endpoint conditions

$$y(a) = y_a, \quad y(b) = y_b,$$

and an additional integral constraint

$$K[y] = \int_a^b G(x, y, y') dx = \ell.$$

The name comes from the classical question of maximising area under a fixed perimeter. In the rope problem the fixed quantity is not a perimeter, but the arc length of the curve.

7.2 The Multiplier Principle for Functionals

Let Y be a candidate constrained extremal, and consider variations

$$y_\varepsilon(x) = Y(x) + \varepsilon g(x), \quad g(a) = g(b) = 0.$$

For an unconstrained problem every such g is admissible. With the constraint $K[y] = \ell$, however, the variation must preserve the constraint to first order:

$$\delta K[Y; g] = 0.$$

Therefore stationarity of J is only required for those directions:

$$\delta J[Y; g] = 0 \quad \text{whenever} \quad \delta K[Y; g] = 0.$$

For fixed Y , the maps

$$g \mapsto \delta J[Y; g], \quad g \mapsto \delta K[Y; g]$$

are linear functionals on the vector space of smooth test functions vanishing at the endpoints. In the regular case, $\delta K[Y; \cdot]$ is not the zero functional. The condition above says that $\delta J[Y; \cdot]$ vanishes on the kernel of $\delta K[Y; \cdot]$. Hence the two linear functionals are proportional:

$$\delta J[Y; g] + \mu \delta K[Y; g] = 0 \quad \text{for all endpoint-preserving } g$$

for some constant $\mu \in \mathbb{R}$.

Theorem 7.1 (Isoperimetric Euler–Lagrange Theorem). *Let Y be a regular extremal of*

$$J[y] = \int_a^b F(x, y, y') dx$$

subject to

$$K[y] = \int_a^b G(x, y, y') dx = \ell$$

and fixed endpoint conditions. Then there exists a constant $\mu \in \mathbb{R}$ such that Y satisfies the Euler–Lagrange equation for the augmented Lagrangian

$$H(x, y, y') = F(x, y, y') + \mu G(x, y, y') :$$

$$\frac{d}{dx} \left(\frac{\partial H}{\partial y'} \right) - \frac{\partial H}{\partial y} = 0.$$

Proof sketch. The multiplier argument gives

$$\delta J[Y; g] + \mu \delta K[Y; g] = 0$$

for every smooth g with $g(a) = g(b) = 0$. By linearity of the first variation, this is

$$\delta(J + \mu K)[Y; g] = 0.$$

Thus Y is stationary for the unconstrained functional

$$\int_a^b \left(F(x, y, y') + \mu G(x, y, y') \right) dx.$$

Applying the ordinary Euler–Lagrange theorem to this augmented integrand gives the stated equation. $\square$

Remark 7.1. The multiplier μ is a constant, not a function of x , because the constraint $K[y] = \ell$ is one scalar equation. If instead one imposed a pointwise constraint at every x , the multiplier would generally be a function.

7.3 The Beltrami Identity

The catenary calculation becomes shorter if we use a first integral of the Euler–Lagrange equation.

Proposition 7.1 (Beltrami Identity). *Suppose*

$$I[y] = \int_a^b H(y, y') dx,$$

so that the integrand has no explicit x -dependence. If Y satisfies the Euler–Lagrange equation for H , then

$$H - Y' H_{y'} = C$$

for some constant C .

Proof. Differentiate $H - Y' H_{y'}$ with respect to x :

$$\frac{d}{dx} (H - Y' H_{y'}) = H_y Y' + H_{y'} Y'' - Y'' H_{y'} - Y' \frac{d}{dx} H_{y'} = Y' \left(H_y - \frac{d}{dx} H_{y'} \right).$$

The last factor vanishes by the Euler–Lagrange equation, so $H - Y' H_{y'}$ is constant. $\square$

7.4 Application: The Catenary

Return to the hanging rope from Section 3.1. We write the unknown curve as $y = y(x)$, fix the endpoints

$$y(a) = y_a, \quad y(b) = y_b,$$

and prescribe the rope length

$$\int_a^b \sqrt{1 + (y')^2} dx = \ell.$$

Ignoring the positive physical constant in the potential energy, the functional to be minimised is

$$J[y] = \int_a^b y \sqrt{1 + (y')^2} dx.$$

Thus

$$F(y, y') = y\sqrt{1 + (y')^2}, \quad G(y, y') = \sqrt{1 + (y')^2}.$$

The multiplier theorem tells us to use

$$H(y, y') = F(y, y') + \mu G(y, y') = (y + \mu)\sqrt{1 + (y')^2},$$

where μ is a constant determined together with the endpoint and length conditions.

Since H does not depend explicitly on x , the Beltrami identity gives

$$H - y'H_{y'} = C.$$

Writing $p = y'$, we have

$$H_p = (y + \mu) \frac{p}{\sqrt{1 + p^2}},$$

and therefore

$$H - y'H_{y'} = (y + \mu)\sqrt{1 + (y')^2} - y'(y + \mu) \frac{y'}{\sqrt{1 + (y')^2}} = \frac{y + \mu}{\sqrt{1 + (y')^2}}.$$

Hence

$$\frac{y + \mu}{\sqrt{1 + (y')^2}} = \alpha$$

for some constant α . For the physical catenary we take $\alpha > 0$. Solving,

$$\sqrt{1 + (y')^2} = \frac{y + \mu}{\alpha},$$

so

$$(y')^2 = \frac{(y + \mu)^2 - \alpha^2}{\alpha^2}.$$

Introduce

$$u = \frac{y + \mu}{\alpha}.$$

Then $y' = \alpha u'$, and the equation becomes

$$1 + \alpha^2(u')^2 = u^2.$$

Thus

$$\frac{du}{\sqrt{u^2 - 1}} = \pm \frac{dx}{\alpha}.$$

After integration,

$$\operatorname{arcosh} u = \pm \frac{x - x_0}{\alpha}.$$

Since $\cosh$ is even, the sign can be absorbed into x_0 , and

$$u = \cosh \left(\frac{x - x_0}{\alpha} \right).$$

Therefore

$$\boxed{y(x) = \alpha \cosh \left(\frac{x - x_0}{\alpha} \right) - \mu}$$

or, after writing the vertical shift as C ,

$$\boxed{y(x) = \alpha \cosh \left(\frac{x - x_0}{\alpha} \right) + C.}$$

This is the **catenary**. The constants α , x_0 , and C are fixed by the two endpoint conditions and the length constraint.

7.5 Practical Recipe

For a single integral constraint, the computational procedure is:

1. Write the target functional $J[y] = \int F(x, y, y') dx$.
2. Write the constraint $K[y] = \int G(x, y, y') dx = \ell$.
3. Check that the constraint is regular, meaning δK is not the zero functional.
4. Form the augmented Lagrangian $H = F + \mu G$ with constant μ .
5. Apply the Euler–Lagrange equation to H .
6. If H has no explicit x -dependence, use the Beltrami identity.
7. Determine the remaining constants from the boundary conditions and the integral constraint.